Meenaly Yadav

+91-9685517616 | 135meenalyyadav@gmail.com | linkedin.com/in/meenaly-yadav

Education

Indian Institute of Science, IISc — **Aug. 2021 – June 2023**
Master of Technology, Computer Science — *Bengaluru, Karnataka*

Malviya National Institute of Technology, NIT — **Aug. 2020 – June 2021**
Master of Technology, Computer Science, 8.25 CGPA — *Jaipur, Rajasthan*

Institute of Engineering and Technology, IET,DAVV — **Aug. 2014 – May 2018**
Bachelor of Engineering, Computer Science, 74.36% — *Indore, Madhya Pradesh*

Relevant Coursework

- Design and Analysis of Algorithm
- Machine Learning and Signal Processing
- Linear Algebra and Probability
- Deep Learning - Natural Language Processing
- Game Theory
- Computer Architecture
- Computational Topology
- Data Analytics

Experience

Microsoft Research India — **Oct 2024 – Current**
Data and Applied Scientist — *Bangalore, Karnataka*

- Leveraged advanced machine learning algorithms and deep learning models to enhance demand forecasting accuracy for Azure Cloud, ensuring optimized resource allocation and inventory management.
- Developed AI-driven solutions for medium-term and short-term cloud capacity planning, addressing dynamic customer needs and reducing latency in supply chain decision-making processes.
- Implemented predictive analytics for customer capacity consumption forecasting, enabling proactive planning and improved service delivery.

Aptiv — **June 2023 – Sept 2024**
Perception — *Bangalore, Karnataka*

- Executed the Validation and Acceptance of the functionality and performance of the TIDL (Texas Instruments Deep Learning) framework for deep learning applications
- Implemented and executed various applications leveraging the TIDL framework.

Samsung Research India — **Sept 2022 – Nov 2022**
Research Intern in 6G Lab — *Bangalore, Karnataka*

- Implemented the Accelerating training and inference of Graph Neural Networks with Fast Sampling and Pipelining.

Accenture — **May 2018 – August 2018**
Associate Software Engineer — *Mumbai, Maharashtra*

- Worked in Accenture as an Application Development Associate.

Projects

Visualization of Higher Dimension Data(Extremum Graph Analysis) *Prof. Vijay Natarajan*
Numpy, scipy, NDDAV — **Aug 2022-June 2023**

- Explored and analyzed high-dimensional functions encountered in scientific data analysis. Enabled interactive exploration of both the topological and the geometric aspect of high-dimensional data.
- Experimented on some higher dimensional dataset by using the NDDAV (N- Dimensional Data Analysis and Visualization)
- Developed the system which is combination of techniques from geometry, machine learning, topology, and visualization and how it can have a significant impact in the various scientific field like inertial confinement fusion, tumor detection.

Question Generation using BERT
BERT, Seq-2-Seq, Python, Keras, Hyperparameter Tuning, GLoVe — **Nov 2022**

- Generated question using state of the art model BERT. Fine-tuned the existing BERT model on the SQuad 1.0 data set and compared it with the existing BLEU scores and other comparison metrics to find the results.
- Tried to extend the model for HLSQG which is implementation of seq-2-seq on top of existing BERT model, which further improves the accuracy of the model.

Question Answering using Language models and Knowledge Graphs(KG) (Paper Implementation)
BERT, GNN, Graph, Python, Neural Network, Hyperparameter, Natural Language Processing — **May 2022**

- Identified relevant knowledge from large KG and perform joint reasoning over the QA context and KG
- Utilized relevance scoring with LLMs to estimate KG node importance within QA contexts, and applied joint reasoning by integrating QA context and KG into a joint graph, updated iteratively via GNN.

Time Series Project
ARIMA, XGBoost, LSTM, matplotlib — **Sept 2021**

- Implemented various EDA techniques to clean time series data and get insights from the data
- Used various Machine Learning Algorithms on the time series like ARIMA, Logistic Regression, and Ensemble models like XGBoost and others and proposed and try to implement LSTM on the Time series data

Credit Card Fraud Detection

Logistic regression,SVM , Python, Pandas, Numpy, Seaborn **May 2021**

- Implemented on the dataset is highly imbalanced and except for the transaction and amount all the columns are hidden.
- Used dataset is higher dimensional so executed PCA to reduce the dimensionality of data and then on the top of that have over sampled our training data during cross-validation, not before.
- Implemented classification Algorithms like Logistic Regression, Decision Tree and SVM.

Technical Skills

Programming Languages:Python, C++, C, Latex, CUDA, Java, CPU, GPU
Libraries: Keras, Numpy, Pandas, NLTK, Sklearn, Pytorch, Tensorflow, cuDNN, MLOps, LLM, RNN, SQL, Gen AI, NLP
Developer Tools: PyCharm, NetBeans DEV, Google Collab, Anaconda, GitHub, LATEX, GCP, Bitbucket

Academic Achievements | Extracurricular

- IISc CSA Student Representative
- Organizer of IISc EECS Research Symposium 2023
- Organizer of IISc Open Day
- Scored 99.32 Percentile in Gate 2021.
- Secured All India Rank 3896 in JEE advanced 2014.
- Qualified First Round of International Math Olympiad with 2nd rank in Indore City.
- Selected as an Indian Oil Scholar against Indian Oil Educational Scholarship Scheme 2012.